

**LONGITUDINAL OUTCOME ASSESSMENT FOR FORENSICS:
DOES PARTICIPATION IN INTERCOLLEGIATE, COMPETITIVE
FORENSICS CONTRIBUTE TO MEASURABLE DIFFERENCES IN
POSITIVE STUDENT OUTCOMES?**

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In an age of educational accountability, the forensics community is and will increasingly be called upon to tell what it seeks to do, how well it accomplishes its goals, and what other effects it has. Surprisingly, there seems little interest in such research at this time. (Anderson, 1974, p. 155)

Participation in intercollegiate, competitive debate teaches critical thinking, research identification, collection, organization and assimilation, evidence evaluation, practical and theoretical argument and counter-argument development, persuasive writing, and intelligent, articulate oral advocacy. In addition to these contextual skills, debate participation boasts an impressive array of non-linear benefits. Debaters are more confident and self-assured. They are poised under pressure and think quickly on their feet. Their research on a variety of resolutions leads to increased understanding of socio-political issues. Developing argumentation supporting both sides of a topic teaches balance, understanding and tolerance for other peoples' views. Weighing evidence results in a higher commitment to ethical and moral argumentation. Policy testing and evaluation leads to a higher sense of social and civic awareness; which, in turn, leads to increased social advocacy and governmental participation. Bold claims, to be sure. The question is are they true?

The vast majority of the initial readership of this article – at the very least – believe that the majority of our forensic students do reap at least most of the aforementioned benefits from their debate participation. We may differ on the specific application and appropriateness of certain skills (such as the utility and eventual transference of speed delivery into the “real world”), but, in general, we agree that debate is a unique educational opportunity that teaches a plethora of direct and indirect skills with overarching educational goals. *We* hold these truths to be self-evident, but do *others*?

Currently, there is an almost complete lack of research that demonstrates the long-term effects of participation in forensics. As a result, administrators who are forced to allocate scarce resources within a very competitive environment are faced with no discernable method for prioritizing allocations based upon measurable results or outcomes. Colleagues are forced to ask program directors how their teaching furthers the department's pedagogic goals. Departmental syllabi discuss educational outcomes, goals and objectives. We evaluate our teaching effectiveness, in large part, based upon how well our testing and formal evaluation process matches and reflects those educational objectives. Why do these formal outcomes, goals, objectives and evaluation techniques not exist in competitive debate *if* it really is an extension of the academic environment? Finally, how does our discipline answer the questions of concerned parents who look at the tremendous commitment of time that active competition in forensics requires, hear the stories of endless road trips weekend after weekend, the missed classes, and occasionally, the justification for a C in algebra, and ask, "how does this contribute to my son or daughter's education?"

What is needed for administrators, fellow academicians and parents is quantitatively based assessment data that tracks, evaluates, and assesses populations of both debate and non-debate students as they matriculate through their college careers looking for any significant differences between the two populations. Outcomes at the end of the longitudinal study period could be directly attributed to the differences in the study populations. Unfortunately, current studies offer only random "snapshots" often based upon anecdotal evidence which cannot be considered valid evidence of the educational and social benefits of forensics participation.

Review of Literature

Literally hundreds of studies over the past nine decades have attempted to examine exactly what competitive debate is, what it should teach and to whom, and what benefits, if any, our student competitors should look forward to as they leave the competitive experience. Of the 682 articles reviewed, only twenty-five suggested that the competitive debate experience was anything less than positive. Hill (1993), Horn & Underberg (1993), and Greenstreet (1993) concluded that empirical evidence to support the claim that competitive debate teaches critical thinking is slight, at best. Husman, Ware and Gruner (1972) found that "debate ability correlates highly with critical thinking regardless of the amount of debate experience" (265). Although they did not draw the conclusion, a possible

inference is that actual experience in debate has little or no effect on critical thinking. As McGlone (1974) posits “ . . . the possibility exists that debate and debate courses attract persons with superior reasoning powers. It may be that critical thinking is a characteristic already possessed by debaters, rather than an effect of debate training” (143). These findings question the ability of traditional behavioral research to correctly identify the direction of the casual relationship. Does debate “cause” behaviors or are those behaviors already present within a population that is naturally attracted to forensics? If the later is the case, then debate does not “cause” anything, but benefits from behaviors and personality types that are already present in our forensics recruits.

Researchers (Colbert, 1994; Infante, Trebing, Shepherd & Seeds, 1984) caution that competitive debaters often exhibit increased verbal aggression and argumentativeness. Hetlinger and Hildreth (1961) concluded that debaters value friendships less and desire to defeat people more than non debaters. Wood & Rowland-Morin (1989), Jones (1994), and Williams, McGee & Worth (2001) warn of the pedagogic and personal tradeoffs when competition becomes the prime motivator for continued participation. Pratt (1990) cautions that debaters learn such a narrow application of specific skills that business professionals do not value those skills. Hunt, Gerard & Simerly (1997) points to the competitive debate student population as academically at-risk. Several studies (Greenstreet, Joekel, Martin, & Piercy 1998; Greenstreet & Frederick, 2000; Jensen, 1994; Loge, 1991; Logue, 1985a, 1985b, 1987a, 1987b; Preston, 1997; Rogers, 1995, 1997; Simerly, Biles & Scott, 1992; Simerly & Gartin, 1994; Stepp, 1991; Stepp, Simerly & Louge, 1993; Stenger & Roth, 1998; Stenger 1999) argue that the competitive playing field is biased against the voices and argument styles of women and minorities and fails to serve “at-risk” populations. However, when all was said and done, even those twenty-five articles concluded that while debate participation might carry with it some ‘risks’ or ‘drawbacks,’ the potential benefits to the students and the programs they represented far outweighed those risks.

On the positive side, competitive debate is credited with increasing critical thinking skills (Beckman, 1957; Brembeck, 1949; Colbert, 1987; Cross, 1961; Gruner, Hussman & Luck, 1971; Horn & Underberg, 1993; Howell, 1943; Husman, Ware & Gruner, 1972; Jackson, 1961; Rowland, 1995; Williams, 1951; Williams et al, 2001), increasing public presentational skills (Colbert & Biggers, 1985; Millsap, 1998; Stenger, 1999; Williams et al, 2001); teaching public advocacy and social responsibility (Bartenan, 1998; Brand 2000; Brownlee, 1978; Derryberry, 1998; Williams et al, 2001) and offering excellent professional

training (Colbert & Biggers, 1985; Hill, 1983; Schnieder, 1984; Spangle & Knapp, 1996). It also increases knowledge, self-confidence, poise, and a wide range of skills necessary for academic success (Bartenan, 1998; Colbert & Biggers, 1985; Derryberry, 1998; Hill, 1983; Jones, 1994; Williams et al, 2001).

Pundits argue that debate teaches social responsibility and advocacy (Bartenan & Frank, 1994; Freely, 1996; Hollihan & Baaske, 1994; Jones, 1994; Rowland, 1995) and enhances a student's academic and professional abilities (Carleton, 1949; Colbert & Biggers, 1985; Derryberry, 1998; Hill, 1983; Jones, 1994; Level, 1957; Pratt, 1990; Schneider, 1984; Spangle & Knapp, Stenger, 1999; 1996; Walker, 1971; Williams, et al 2001). In addition, researchers (Bartenan, 1998; Derryberry, 1998; Millsap, 1998) have concluded that debate teaches important leadership skills. As Bartenan, (1998, p. 12-13) concludes: "debate fosters leadership skills of reflection, connectedness and advocacy. Forensics programs are valuable models of learner-centered pedagogy and underutilized resources for diversity education on the liberal arts campus." Susan Millsap (1998) goes so far as to say that a forensics education alone has the ability to be pedagogically effective across the liberal arts curriculum. She believes that instructors in other areas are not giving their students training in the skills necessary to make their pedagogy effective. Perhaps, Daniel Rohrer (1987) summarizes the pedagogic virtues of a debate education best when he quotes Ralph Waldo Emerson as saying "The highest purpose of debate is to develop 'man thinking,' in the total milieu of society and the world around him. Ultimately, debate attempts to improve a man by laying a foundation for better understanding of himself and those around him, to inculcate habits of the mind, breadth of interest, and enlargement of spirit, the same goals of the liberal education" (p. 9).

It would be easy for the reader to conclude at this point, that an experientially-based education in competitive debate is *the* superior model for the advancement and understanding of a host of pedagogic goals within a liberal arts based higher education. However, a more critical analysis of the work presented would highlight a major weakness in the research literature which supports debate as underpinning and reinforcing a pedagogic set of outcomes: an almost complete lack of behavioral research grounded in longitudinal, empirical support. The vast majority of the published research relies on anecdotal examples, or quasi-statistical analysis based upon tiny, even personal samples which are not valid for comparisons to more general populations. Early critics, (Baird 1950; Anderson, 1974; McGlone 1974) warn of this weakness. They even suggest rigorous models with which to

develop credible forensic research that would be valid from a behavioral perspective. Sadly, little, if any, significant behavioral research has been conducted over the past three decades.

The real question we must investigate as a discipline is whether competitive debate teaches beyond personal testimonies and theories. Thus far, we have the personal observations of literally hundreds of years of coaching experience to support the argument that debate serves its students and the broader communities of higher education and society, in general. However, without valid longitudinal studies to compare those students who participate in forensics with those who do not over an extended period of time to investigate and record measurable differences, valid conclusions regarding the benefits of the experience cannot be advanced.

Method

Research Questions. An analysis of the positive behavioral themes within the literature led to the collection of 56 positive outcomes or claims linked to debate participation. Researchers (Babbie, 1992; Wimmer & Dominick, 2000) advance the use of focus groups to narrow both the vagueness and numbers of variables. In this case, since many of the positive outcomes seemed to be very similar, the focus group method was desirable. Three focus groups were assembled. Each group consisted of administrators (two with previous debate experience – one former DOF, one former college debater), program directors, graduate and undergraduate students. Each group was asked to prioritize the 56 positive outcome claims by ranking the top twenty claims in terms of the strength of the efficacy of the research. The groups were then asked to assemble those statements into “like” concepts and to suggest labels for those concept areas. The top 20 claims were categorized into five critical areas: 1) academic success; 2) social responsibility; 3) psychological adjustment; 4) cultural tolerance; and 5) moral/ethical commitment. Once these five critical areas were identified, the focus groups submitted questions designed to measure these positive outcomes through behavioral objectives. Validity of the survey instrument and a complete discussion of the five critical behavioral outcomes are advanced below and in the results and discussion sections.

To that end,

RQ1: Are there significant differences between debate and non-debate student populations in five critical areas: academic success; social responsibility;

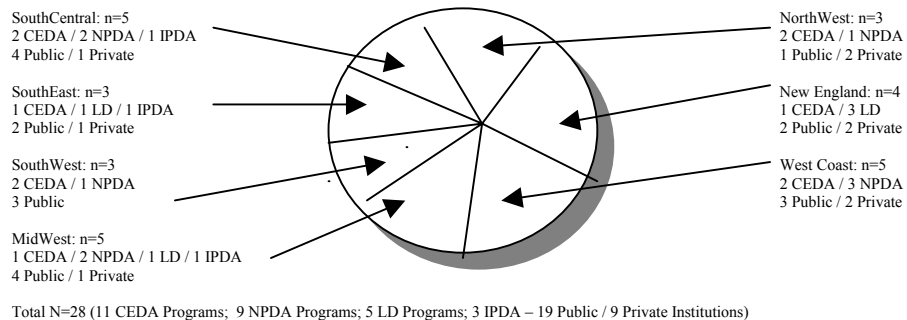
psychological adjustment; cultural tolerance; and moral/ethical commitment?

And if so,

RQ2: Are those differences positive for the debate population?

Subject Selection: In July of 1997, twenty-eight Directors of Forensics were identified and approached to participate in this study. Their selection was based upon the following subjective inclusion criteria: 1) *Representativeness* of differing debate organizations and competitive formats (e.g. CEDA, NPDA, LD and IPDA - Public); 2) *Program profile* in terms of competitive success, size of team, budget and opportunity to travel; 3) *Location* and *Classification* (every attempt was made to derive an inclusive sample of geographical regions and both public and private institutions of higher learning); and 4) *Stability* with regard to the Director of Forensics. Obviously, the nature of a longitudinal study demands continued access to the subject population. Too often, the loss of a director of forensics results in the loss of the program. At the very least, the program may lose its momentum and students may quit due to the transition in coaching style and/or focus. The application of these selection criteria resulted in the following distributed sample:

Table 1
Graphic Representation of Sample Distribution



Subjects. Each Director of Forensics was asked to identify incoming first year students recruited for the Debate Team and twenty incoming first year student volunteers from their basic teaching load with no previous debate experience or expressed interest in joining the host DOF's debate team. All students were briefed on the intent, scope, length and design of the study and signed the informed consent / privacy statements. No participant was

informed that the specific purpose of the research was to measure behavioral outcomes linked to forensics participation. It must be conceded that there are literally hundreds of variables which might impact the outcome of this type of study. Obviously, there are some concerns with an initial sample based upon the randomness of who might join a program and the convenience of who might be in the program directors' class loads and willing to volunteer. Caution regarding the potential bias of the program directors that made the initial selections and the possibility of variance in demographics, life experience, and attitudes of the subjects should not be ignored. A basic intake survey was distributed that attempted to address any significant differences between the two groups of student participants which addressed demographics, high school academic, extracurricular and social backgrounds.

760 incoming first-year college students were included in the original survey group (563 non-debate and 197 debate). Data was collated and manipulated keeping in mind the original parameters of the inclusion criteria. Based on the intake survey, comparison groups of debaters and non-debaters were constructed based on the goal of minimizing all demographic, academic, extracurricular and social differences between the two groups. Of the original pool, 100 debate and 100 non-debate students were selected for participation in the four-year study. The comparison of the study groups is reported in table 2.

Table 2
Study Group Comparisons

Participant Group:	Gender		Ethnicity					High School		ACT	SAT	GPA	SocioEconomic ¹
	Male	Female	White	Black	Asian	Hispanic	Other	Public	Private				
Debate n=100	63	37	71	9	12	7	1	75	25	23.6	1126	3.14	\$ 63,587.00
Non-Debate N=100	60	40	70	13	10	5	2	73	27	24.1	1120	3.21	\$ 69,117.00
	Major						Hours Pursued	Employment			Scholarship/Financial Aid ³		
	Liberal Arts	Science	Business	Pre-Professional ²				Part-time	>Part-time		Partial	Half	Full
Debate N=100	28	17	11	34			17	41	7		47	71	88
Non-Debate N=100	37	22	24	17			18	53	11		53	73	80

Notes:

¹ Average yearly household income;

² Pre-Law, Criminal Justice, Pre-Med;

³ Reported as total numbers – read as 47 received at least partial scholarships or financial aid; 71 received at least half-time (of which those 47 would be included); 88 received full-time financial assistance (of which the 71 would be included).

The exclusion of subjects and the longitudinal design of the study should help to mitigate the inherent weakness of this type of research. The traditional difficulty in interpreting this type of research is the question of whether differences are effects or causes. Do they result from the debate experience (debate enhances critical thinking), or do they merely predict who debates (critical thinkers like to debate)? No research that attempts to measure life-experience outcomes at the high school or collegiate level can ignore the potential impacts to validity. Without tracking children long before they make the “debate” choice, it would be impossible to make sweeping claims on behalf of debate. In this case, caution is recommended and should limit the conclusions drawn.

Survey Instrument. After the conclusion of the first study year (May, 1998), the comparison survey designed to collect self-report data and measure any differences between the debate and non-debate groups in five critical areas: 1) social responsibility; 2) cultural understanding and tolerance; 3) academic success; 4) moral and ethical issues; and 5) psychological multipliers was distributed to the remote testing sites (28 DOF locations). After collection, compilation and recording of the first year data set, the first level of statistical analysis was to assess the validity of the survey instrument. This first level of analysis was conducted in two steps:

Step 1) The data set was subjected to a principle components factor analysis to derive Eigenvalues. Factors with an Eigenvalue of greater than 1.0 were retained as an independent factor by the MINEIGEN program. The amount of variance explained by each factor can be estimated by a relative comparison of the Eigenvalues in table 3.

Table 3
Eigenvalues and Proportion of Variance for Factors

	Factor 1 Academic Success	Factor 2 Social Responsibility	Factor 3 Psychological Adjustment	Factor 4 Cultural Tolerance & Understanding	Factor 5 Moral / Ethical Issues
Eigenvalues	5.017	4.001	3.119	2.798	2.517
Proportion	0.379	0.211	0.181	0.118	0.101
Cumulative	0.391	0.611	0.767	0.899	0.971

5 Independent factors retained (EV>1.0)

Proportion reflects variance accounted for by the factor solution

The MIENIGEN method retains all factors above the inclusion criteria and attempts a five-factor solution using principle components analysis. In this case, all five critical areas

were retained as independent factors each accounting for a portion of the This confirms that each of the five critical outcomes is functioning as an independent measure of variance.

Step 2) After the Eigenvalues were derived, the five-factor solution was confirmed by using orthogonal factor analysis with varimax rotation and then subjecting those factors to ordinary least squares confirmatory factor analysis as described in the work of Hunter and Cohen (1969). Again, all five factors were retained as independent variables. This second level of analysis was conducted to address concerns that some of the five critical outcome areas might be expressed in interdependent relationships. For example, does one necessarily need to demonstrate a high level of moral and ethical thinking to express positive outcomes in social responsibility? Is social responsibility linked to cultural understanding and tolerance? If the critical outcomes were exhibiting interdependent relationships, it would be difficult to isolate variance between the subject groups. An analysis of Table 4 strongly confirms the independence of the five critical outcomes.

Table 4
Principle Components Factor Analysis for Critical Outcomes

Critical Outcomes	Factor 1	Rotated Factor Pattern		Factor 4	Factor 5
		Factor 2	Factor 3		
Academic Success	<i>+0.772</i>	-0.221	-0.317	+0.098	+0.111
Social Responsibility	-0.119	<i>+0.542</i>	+0.111	-0.035	-0.046
Personality Adjustment	-0.217	-0.301	<i>+0.512</i>	+0.118	-0.213
Cultural Tolerance	-0.091	-0.107	-0.198	<i>+0.389</i>	+0.190
Moral / Ethical	+0.116	-0.119	-0.189	-0.209	<i>+0.471</i>

All factor loadings above the .35 level are considered significant and meet the inclusion criteria and are expressed in bold italicized type.

Data Collection. Since the outcome of the first level of analysis effectively advanced all five critical outcomes as independent variables, the 84-item survey instrument using a standard 7-point scale system of scoring was validated. At the conclusion of each “study year” (May 1998, ‘99, ‘00, ‘01), data was collected, collated and stored. Natural attrition over the course of the study resulted in a final N of 154 students for the complete four-year comparison (debate (n=73) and non-debate (n=81) students). All data from mid-study dropouts were excluded from the final analysis. Eleven of the debate students dropped out of the study by quitting their debate teams but continued to matriculate through their course of studies. Thus, the actual dropout rate for the original study group of 200 was approximately 17%.

At the conclusion of the fourth year, all data were collated and analyzed. For verification of validity, step one of the original statistical analysis was repeated.

Eigenvalues were derived and a principle components factor analysis was performed. No significant differences between the output product of study year one and the completed four-year project were noted. Again, since all five critical outcomes were functioning as independent measures, the logical extension of the research analysis would be: 1) are there significant differences between study groups in the five critical areas; and 2) if so, are those differences positive for forensics participants?

To answer these questions, the data set was divided into two groups: *debate* and *non-debate*. Once divided the scores in each of the five critical outcome areas were averaged and compared using paired *t-tests*, which is similar to a one-sample *t-test* on differences ($H_0: d=0$ v $H_a \neq 0$). The higher the number of comparisons made, the greater the risk of a Type 1 error. To protect the integrity of the process, Bonferroni's approach to multiple comparisons was used (.05 divided by 2 times the number of comparisons (154) = $\text{Prob} > |T| = .0002$). As a result, any comparison where $p < .0002$ was considered statistically significant. SPSS was used to analyze and manipulate the data set. Those results are reported in table 5.

Table 5
Response Comparisons for Statistical Significance

	Critical Outcomes				
	Factor 1 Academic Success	Factor 2 Social Responsibility	Factor 3 Psychological Adjustment	Factor 4 Cultural Tolerance & Understanding	Factor 5 Moral / Ethical Issues
Respondent Group					
Debate	+1.675	+1.315	+1.876	+0.963	+0.539
Prob > T	0.0001	0.0001	0.0001	0.0001	0.0001
Non-debate	+0.997	+0.436	+0.195	+0.651	+0.139
Prob > T	0.0001	0.0001	0.0001	0.0001	0.0001

All values represent means. Positive values equal positive relationships.

While Table 5 does not report correlated relationships, but simply significant differences in the way the subject groups responded to the statements measuring the five critical outcome areas, the obvious differences in perception are interesting. At the very least, the two respondent groups perceive of themselves very differently. These perceptual differences are easier to understand in graphic representations. For ease of access, all tables are reported in percentages with significance data listed below. Only those relationships between debate and non-debate subject groups which reflected significant differences are

included. Correlated coefficients for the subject groups are entered as a basis of comparison at the bottom of each graph. The final step in this statistical analysis was to analyze the significant differences for the relative strength of the correlations. *Pearson's r* correlational formula was used to derive coefficients. All Pearson Correlation Coefficients are expressed as $\sqrt{r^2}$ under $H_0: \rho=0$ (N154).

Results & Discussion

Since each of the five critical outcome areas was validated as having an independent relationship to the subject groups, each outcome was examined independently and assessed in terms of its impact on the subject group.

The first critical outcome analyzed was *social responsibility*. A cluster of 17 questions was asked under the social responsibility matrix of the instrument. These questions were separated into 4 distinct measures targeted towards how the subject groups responded through direct involvement in: **1) voting** - defined as casting a vote in a local, state or national election in at least 50% of the elections for which the respondent was eligible; **2) social volunteerism** - defined as volunteering more than 5 hours per week in a social organization outside of the scope of their university studies or normal social intercourse (e.g. some listed organizations were community services, scouting, church work, Big Brothers / Big Sisters, nursing homes, animal shelters, Greenpeace, various ecological preservation societies, museums, and library reading programs – *debate and/or mandatory participation tied to the university service component were excluded*); **3) political volunteerism** – defined as working more than five hours per week in a political campaign, party, or involved in campus Democrats or campus Republicans and/or the student government association; **4) participation in social advocacy** – defined as active participation designed to bring about social change (e.g. fund-raising, speaking, writing, or serving in not-for-profits aimed at social services, public policy, charities and/or religious organizations). The results of those comparisons are reported in Table 6.

Within the critical outcome area of *social responsibility*, *debate* respondents experienced much higher positive outcomes than their *non-debate* peers. All four comparisons were statistically significant. Debate students were much more likely to consistently vote in political elections both on campus and in the local, state and national elections. The positive correlation between debate and political volunteerism was very strong. Over three-quarters of the debate students were directly involved in specific political campaigns, political party work, and/or political student organizations. Additionally, debate students were more likely to hold positions of leadership in those organizations

(Mean over 4 years = .43%). Finally, debaters were moderately more likely to be involved in social advocacy; that is, using their speaking / persuasive skills in direct attempts to influence social activism and policy outcomes.

Table 6
Comparison of Social Responsibility Outcomes

Figure 1
Propensity to Vote

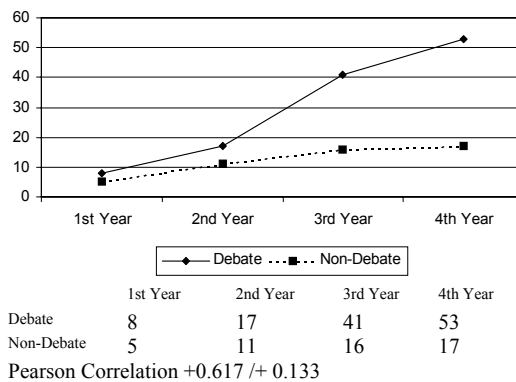


Figure 3
Propensity towards Political Volunteerism

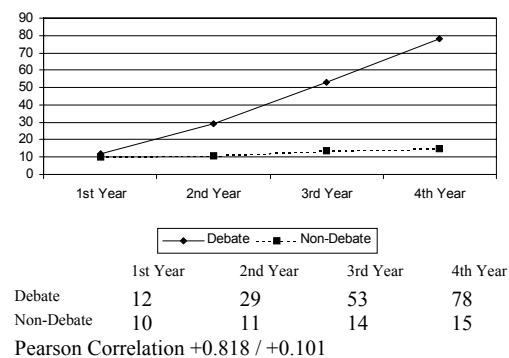


Figure 2
Propensity toward Social Volunteerism

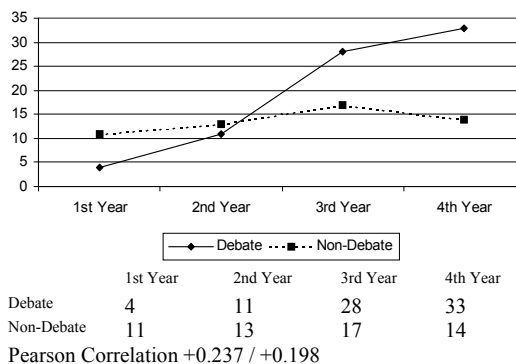
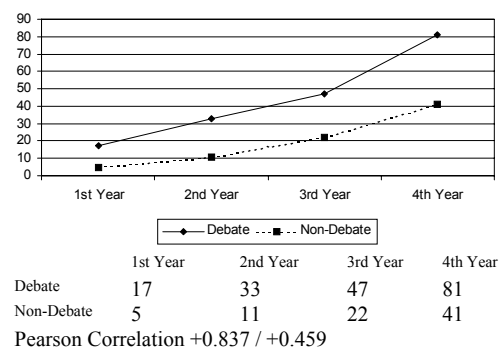


Figure 4
Propensity for Participation in Social Advocacy



Within the outcome area of *Social Volunteerism*, debate students were more likely to respond positively in their third and fourth survey year. Some students cited the initial time commitments of making the transition to college work and the additional challenge of preparation for debate competitions. Once these two time commitments were balanced, the

incidence of social volunteerism increased dramatically for debate students, while the non-debate involvement rates remained relatively static. This data might suggest that given the time, debaters would prefer to devote their time in political involvement and advocacy. As they learn time management skills, social volunteerism increases with rates almost doubling each survey year among the debate respondent group.

The second critical outcome analyzed was *cultural understanding and tolerance*. Eighteen questions were asked to examine differences between the subject groups in four specific areas: **1) enrollment in Cross-Cultural coursework** – defined as elective coursework specifically designed to educate the student towards a deeper understanding of another culture or social identity (e.g. gender, religious, ethnicity, geopolitical). Coursework required to satisfy general education graduation requirements or specific major or minor requirements were not included; **2) Cross-cultural relationships** – defined as a significant interpersonal relationship with a person representative of a different gender orientation, ethnicity, religion, political affiliation, or international culture. To qualify as significant, the respondents were directed to report relationships which had been maintained for more than twelve months; **3) Cross-cultural organization** – defined as membership in an organization with either significant representation among the membership of differing gender orientations, ethnicity, political or religious affiliations, or international cultures, or with a mission statement which was specifically oriented towards diverse cultural groups (e.g. listed organizations included the black caucus; international student club; gay and lesbian caucus; student ecumenical council. Again, the debate team was excluded); and **4) Rejection of Social Norming** – this cluster of seven questions focused on the respondents' propensity to reject society's absolute right to determine "correct" behavior for its citizens. In effect, does the student have the sophistication to understand differing attitudes towards social norming?

Within the critical outcome of *Cultural Understanding and Tolerance*, only two of the four areas were identified as statistically significant in terms of positive outcomes. Debate students were slightly more likely to enroll in cross-cultural coursework. The most significant positive difference noted between respondent groups was within the outcome of *Social Norming*. Debate students were much more likely to reject classical definitions of social norming. There is a very well defined, positive growth curve for this outcome expressed in Figure 2. As debaters become exposed to various resolutions and topics for debate, conduct research on both sides of a usually controversial social subject, organize and

write briefs for both sides, and go through the process of arguing those positions, they have the opportunity to develop a wider view of differing social perceptions. It could be argued that the beginning of increased social tolerance is the rejection of what the dominant culture considers “normal;” and thus, acceptable. As our definition of “normal” widens, we become more socially tolerant of previously “taboo” outlooks and perceptions.

Table 7
Comparison of Cultural Tolerance and Understanding Outcomes

Figure 1
Propensity to Enroll in Cross-Cultural Coursework

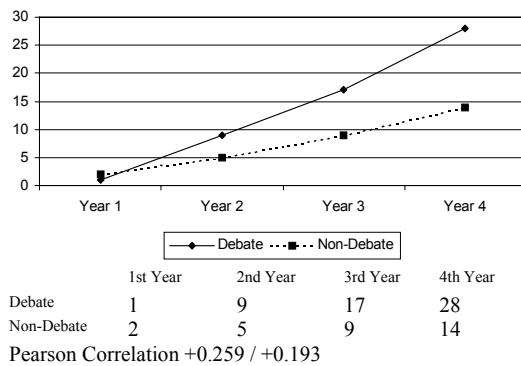
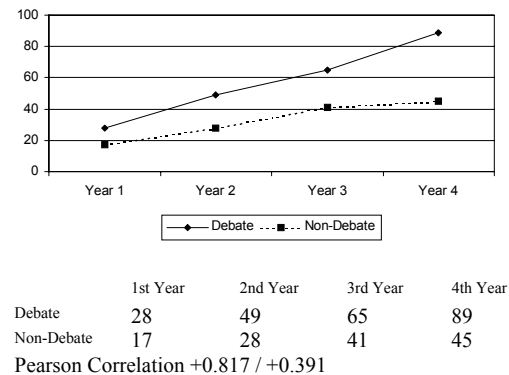


Figure 2
Propensity to Reject Classical Norms

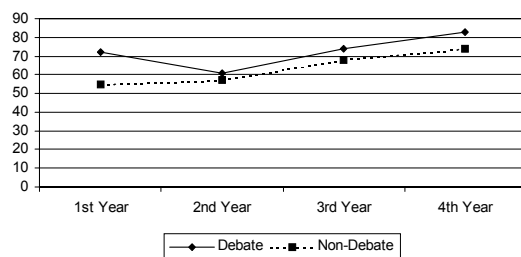


Comparisons of *Academic Success* were examined third. This critical outcome area was perhaps the most straightforward area of comparison. Six measures were advanced in 9 questions: **1) GPA** – defined as the ability to maintain a GPA of 3.50 or above in a 4.0 system; **2) Propensity to Matriculate on Time** – defined as having successfully completed the requisite hours to maintain satisfactory academic progress towards a four-year graduation schedule. Summer sessions were considered a part of the preceding academic year; **3) Propensity to Change Majors** – defined as a change in major that would require a change of academic departments and advisors; **4) Professional Internship Involvement** – defined as an external, professional internship directed towards their major area of studies; **5) Acceptance into Graduate Schools / Programs**; and **6) Job Offers** within major course of study at time of graduation.

An analysis of Table 8 should be of particular interest to experienced DOF's more so for the common myths it *may* dispel rather than any significant differences of note. Figures 1 through 3 note no significant differences between subject groups in 1-GPA, 2- the ability to matriculate on time or 3-the propensity to change majors. Debaters' GPA's started slightly higher than the non-debate group and remained slightly higher throughout the study (Debate graduation average GPA: 3.57 / non-debate: 3.39). Debaters' tendency to matriculate on time slowed during their sophomore and junior years, but no significant differences were noted when compared to

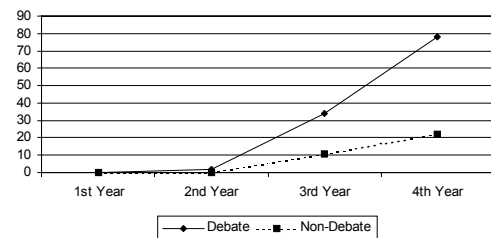
Table 8
Comparisons of Academic Success

Figure 1
Propensity to Maintain GPA above 3.5 (4.0 scale)



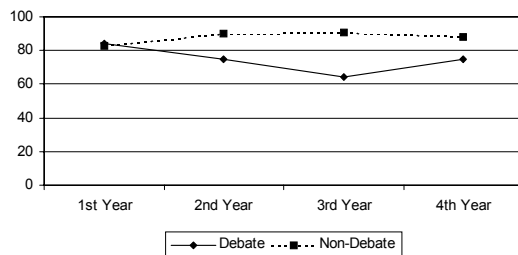
Pearson Correlation +0.753 / +0.597

Figure 3
Propensity for Professional Internship Involvement



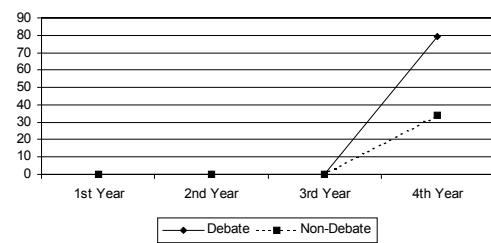
Pearson Correlation +0.801 / +0.253

Figure 2
Propensity to Matriculate on Time



Pearson Correlation: No significant differences

Figure 4
Acceptance into Graduate School Programs



Pearson Correlation +0.715 / +0.261

Table 8
Comparisons of Academic Success (Cont.)

Figure 5
Propensity to Change Major

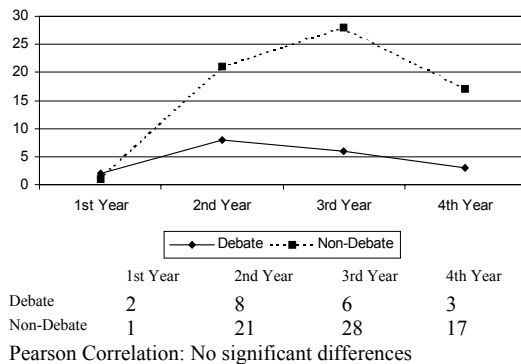
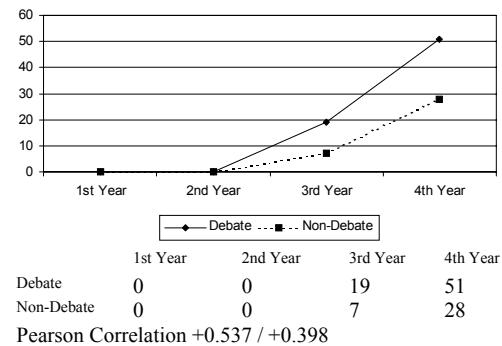


Figure 6
Job Offers in Major at time of Graduation



non-debate students, which resulted in very similar on-time graduation rates for both study groups. According to Figure 3, debaters were also somewhat less likely to change their majors. In summary, any inference that involvement in debate significantly lowers GPA's, increases matriculation time, lowers graduation rates, or leads to indecisiveness in choice of major was not statistically supported in this study. Debate experience did not result in negative outcomes for academic success within the study groups.

In contrast, Figures 4 – 6 record significant positive outcomes. There is a strong correlation between debate experience and involvement in professional internships, which support the student's major course of study. This involvement may be linked to a moderately high correlation between debate and job offers at date of graduation rates reported in Figure 6. Finally, Figure 5 notes a strong correlation between debate experience and acceptance into graduate programs and professional schools. In conclusion, with regard to the third critical outcome of *Academic Success*, debate experience has definite positive pre and post-graduation, "real world" outcomes. Just as important, the negative perceptions of lower GPA's, slower matriculation and graduation, and unfocused wandering from major to major are not statistically supported for the debate study population.

The fourth critical outcome examined was a comparison of the **Ethical / Moral** dimension of the respondent groups. Fifteen survey items were designed to examine four

areas of comparison: **1) Propensity to Distort the Truth** – defined as the respondents' comfort with truth distortion (e.g. How likely are you to deliberately lie to advance your position so long as it hurts no one else?; Would you distort facts or information to advance your position so long as you knew that you were right?); **2) Belief in Situational Ethics** – defined as the respondents' comfort with shifting ethical standards dependent upon the situation (e.g. Do circumstances sometimes justify actions that would otherwise be considered wrong or immoral?); **3) Propensity to Ignore Conflicting Evidence** – this area is directed as the students' propensity to ignore evidence which might contradict their position. How open-minded are they? Are the students willing to investigate balanced discussions on controversial content areas and engage in critical thinking skills to make decisions?; **4) Belief in the Just Society Tradition** – defined as the comfort level of respondents' with society's system of justice. Can social advocates make meaningful significant changes?

No significant differences were reported between the subject groups in the areas of propensity to distort the truth or a belief in situational ethics. Again, the data is perhaps more interesting for what it does not report. Any inference that debaters are more likely than their peers to distort the truth or to resort to situational ethics to “win the point” received no statistical support in this study.

Table 9
Comparison of Moral / Ethical Outcomes

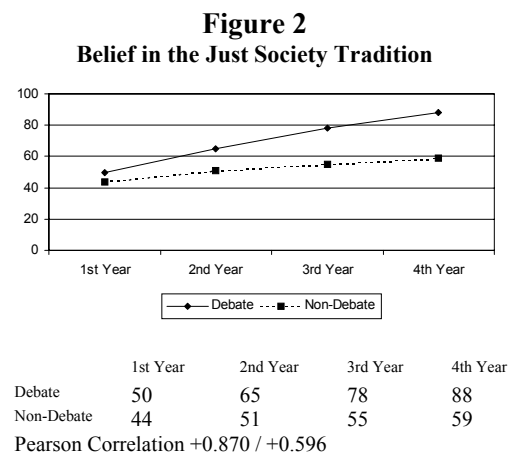
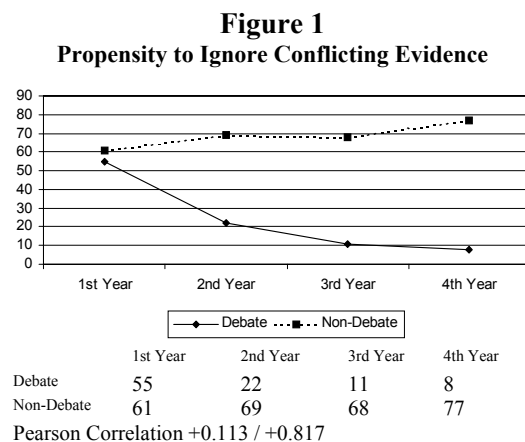


Table 9 reflects an overall positive outcome assessment for debate experience with regard to propensity to ignore conflicting evidence and a belief in a just society tradition. According to Figures 1 and 2, debaters are significantly less likely to ignore evidence which conflicts with what they believe and are significantly more comfortable with society's system of justice. There appears to be a stronger belief in the ability of concerned citizens to make a difference on the part of the debate study group when compared to their non-debate peers.

The fifth and final area of critical outcome analysis was conducted on **Psychological Adjustment**. This area is conceptualized as psychological indicators of behaviors that would assist students towards a healthy, positive outlook; which would in turn lead to a more successful college experience. Twenty-five questions were asked and grouped into six measurement areas: **1) Feelings of Depression / Anxiety** – intended as an indication of attitude, not as a chronic diagnosis; Do you feel depressed; and if so, how many times per week do you feel depressed? Do you feel anxious in social, academic, or work environments? Do you ever feel as though you want to give up? **2) Overwhelmed under pressure** – academic, work or social; Difficulty in concentration? Failure to organize, focus or complete assignments? **3) Feelings of Self-Confidence** – general measure of the respondent's sense of confidence and positive outlook; **4) Confidence in Communication Skills** – how comfortable is the respondent with their ability to successfully communicate their thoughts, feelings and positions in oral and written form?; **5) Ability to Maintain Long-term Relationships** – defined as a significant friendship or romantic interest of more than 12 months in duration; and **6) Propensity for Flexibility** – How flexible is the respondent's outlook on life, social issues and situations? Does lifestyle dictate life choices? Are they able to adapt to changes in schedule?

In terms of significant differences, Table 10 records five outcomes for those who participate in debate: 1) debaters report significantly less feelings of depression and anxiety; 2) significantly less feelings of being overwhelmed under pressure; 3) significantly higher feelings of confidence and positive outlook; 4) significantly higher confidence in their ability to communicate their feelings and positions through oral and written communication; and 5) significantly more flexible. The only area where the study groups exhibited no significant differences was in their ability to maintain successful long term relationships. Again, this finding is probably more interesting in what it does not report. Any inference that debate subjects found it more or less difficult to sustain interpersonal relationships than

their non-debate peers. Differences in the critical area of Psychological Adjustment are reported in Table 10.

Perhaps the best way to answer the questions RQ1: “Are there significant differences between debate and non-debate student populations in five critical areas: academic success; social responsibility; psychological adjustment; cultural tolerance; and moral/ethical commitment?” And if so, RQ2: “Are those differences positive for the debate population?” is to chart the data and let it speak for itself. Those results are reported in table 11.

Table 10
Comparisons of Psychological Adjustment

Figure 1
Feelings of Depression/Anxiety

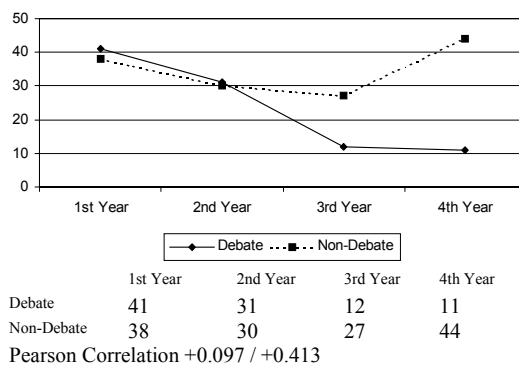


Figure 3
Feelings of Self Confidence/Positive Outlook

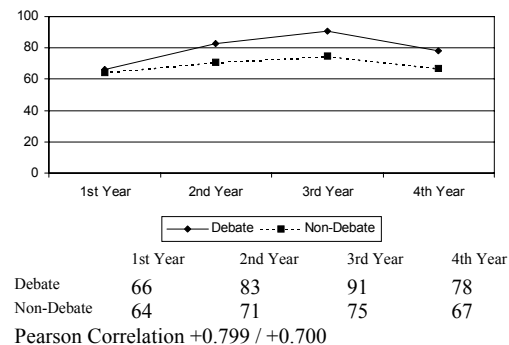


Figure 2
Feelings of being Overwhelmed under Pressure

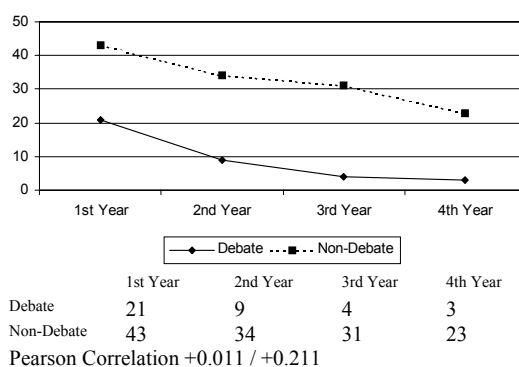


Figure 4
Confidence in Communication Skills

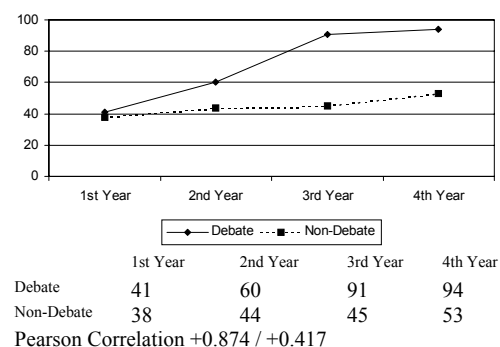


Table 10
Comparisons of Psychological Adjustment (Cont.)

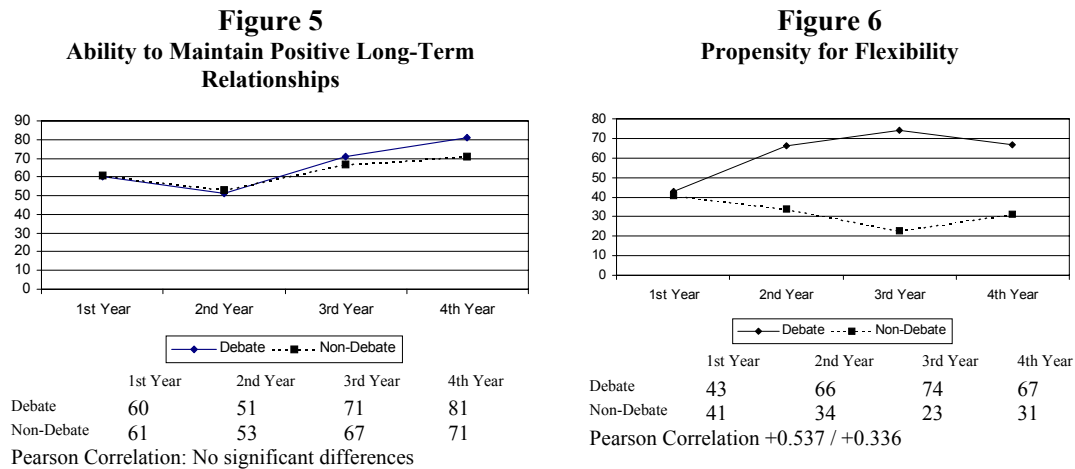


Table 11
Consolidated Comparison between Debate and Non-Debate Groups

	Study Populations	
	Debate Students	Non-debate Students
Five Critical Outcomes		
Social Responsibility		
Propensity to vote	+0.617	+0.113
Propensity towards Social Volunteerism	+0.237	+0.198
Propensity towards Political Volunteerism	+0.818	+0.101
Propensity for Participation in Social Activism	+0.837	+0.459
Cultural Tolerance and Understanding		
Enrollment in Cross-Cultural Coursework	+0.259	+0.193
Maintain Cross-cultural relationships	No significant difference	
Involvement in Cross-Cultural Organizations	No significant difference	
Reject Classical definition of social norming	+0.817	+0.391
Academic Success		
Propensity to maintain 3.5 GPA	+0.753	+0.697
Propensity to Matriculate on time	No significant difference	
Propensity to change major	No significant difference	
Propensity for Professional Internship	+0.081	+0.253
Acceptance to Graduate Program	+0.715	+0.261
Job Offers in Major at Graduation	+0.537	+0.398
Moral / Ethical Outcomes		
Propensity to Distort the Truth	No significant Difference	
Belief in Situational Ethics	No significant difference	
Propensity to Ignore Conflicting Evidence	+0.113	+0.817
Belief in the Just Society Tradition	+0.870	+0.596
Psychological Multipliers		
Propensity for Depression / Anxiety	+0.097	+0.413
Feeling Overwhelmed under Pressure	+0.011	+0.211
Feelings of Self-Confidence / Positive Outlook	+0.799	+0.700
Confidence in Communication Skills	+0.873	+0.417
Maintain long-term relationships	No significant difference	
Propensity for Flexibility	+0.537	+0.336

Note: All positive behavioral comparisons appear in bold face.

There are significant differences between the debate and non-debate student populations in 17 of the 24 focus areas defined as critical outcomes. The answer to the second question regarding whether those differences were positive outcomes for the debate population can also be clearly inferred from the data analysis. Within this study population, at the end of their collegiate tenure, debate participants evidenced the following positive traits:

Social Responsibility

Debaters were significantly more likely to participate in the democratic process through voting, volunteering their time and resources to political campaigns, and participating in social activism. They were also slightly more likely to participate in social volunteerism.

Cultural Tolerance and Understanding

Debaters were more likely to enroll in cross-cultural coursework not tied to their major, minor or general education requirements. Debaters were much more likely to exhibit tolerance for and acceptance of the views and contributions of other races, religions, cultures, and issues of sexual preference, through the rejection of classical definitions of social norming.

Academic Success

Debaters were able to maintain slightly better GPA's than their non-debate peers. They were significantly stronger academically through much higher rates of professional internships while in college, acceptance to graduate programs, and job offers within their chosen majors at graduation. There were no significant differences regarding graduating on time and maintaining their focus of major.

Moral / Ethical Outcomes

Debaters were significantly more likely to believe in the just society tradition (ethics, values, appropriate social contracts) than their peers. They were also much less likely to ignore the impact of conflicting evidence to their viewpoints and beliefs – a factor often identified as critical to the ability to think and reason critically. No significant differences were reported in the likelihood to distort or manipulate the truth or believe in situational ethics.

Psychological Adjustment

Debaters were much less likely to experience feelings of anxiety, depression, or being overwhelmed when under pressure. They exhibited stronger, more positive outlooks and feelings of self-confidence. They were significantly stronger in their confidence in their communication skills to evaluate and clearly articulate their position through oral and written channels. No significant differences were reported with regard to the ability to maintain long-term relationships.

Conclusions

The conclusions for this study population are clear: in almost every case, in almost every area examined, participation in debate had significant positive outcomes. Even in those areas where no significant differences were found, those results were not necessarily negative. Participation in debate was not shown to significantly impact debaters' ability to graduate on time, stick with their major, maintain significant long-term interpersonal relationships, respect the truth, or become involved in cross-cultural relationships or activities. To the contrary, students engaged in debate participation *seem* to have adjusted to the social and interpersonal aspects of college life without significant exception when compared to their non-debate peers. It could be inferred that the stereotype of the maladjusted, debate nerd pining away at the extreme edge of college life is unfounded. Debaters are often better adjusted than their non-debate peers with lower rates of depression, anxiety, and feelings of being overwhelmed under pressure in addition to higher feelings of self-confidence in both their social and academic abilities.

The positive outcomes of debate participation are overwhelming for the study group, and include greater political and social awareness and participation; an increased awareness of and tolerance for intercultural differences; increased involvement in professional internships, acceptance to graduate programs, job offers at graduation; a deeper understanding and respect for ethics and the proper evaluation of evidence; and, stronger, healthier personality profiles. Debate participation, in this case, was significantly correlated with positive outcomes.

Of course, some care should be taken when making generalized claims from this study sample to the larger debate and non-debate populations across the United States. Further research with larger study populations is absolutely critical to the proper evaluation of the

claims explored through this study. It would also be interesting to follow the populations “post graduation” to see if the positive outcomes reported were long lasting or either dissipated or moderated with time. Do the outcomes of debate participation have lifelong results that continue to engender overall success? Additionally, further statistical analysis and modification of the survey instrument would tighten the conclusions for future efforts. Finally, threats to casual conclusions should be explicitly considered before making more generalized conclusions. This difficulty in interpreting the direction of the casual relationships is somewhat mitigated by the longitudinal nature of the study. However, generalized conclusions regarding the casual relationship between debate participation and positive outcomes should be advanced with caution.

In conclusion, this study should prove useful to competitive debate advocates who require empirical data to persuade administrators, colleagues and parents that participation in intercollegiate debate is beneficial to the student in both the socio-psychological and academic success arenas.

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